Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	*(a)*	(i)		121 kJ mol^{-1}		1		1		
		(ii)		$\Delta H = -(-728) - (242) - (1735) - (178) + (-795)$ (1) $\Delta H = -2222$ kJ mol^{-1} (1)		2		2	1	
		(iii)		award (1) for any value in range 900-1600 kJ mol^{-1} award (1) for either of following • ionisation energies increase for successive ionisation energies (so value must be more than half the ionisation of the first two electrons) • second ionisation energy is greater than first ionisation energy		1	1	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		at minimum temperature $\Delta G = 0$ so equation gives $\Delta S = \Delta H/T$ (1) change units giving T = 473 K and 78000 J mol^{-1} (1) ΔS = 165 J K^{-1} mol^{-1} (allow 164.9) (1) ECF possible	1	1 1		3	3	
		(ii)		0.0284 mol of $CaCl_2$ (1) 0.1138 mol of water (1) x = 4 (1) ECF possible		3		3	2	
		(iii)		brick-red	1			1		1
	(c)			$CaCl_2$ misty fumes (and white solid) (1) $CaBr_2$ will produce orange fumes / solution as well as misty fumes (and white solid) (1) accept will **also** produce orange fumes if first mark awarded bromide is easier to oxidise than chloride (so can be oxidised by the sulfuric acid) (1)	3			3		1 1
				Question 9 total	**5**	**9**	**1**	**15**	**6**	**3**